Horn-Geometric Axioms for Faithfully Quadratic Rings

Abstract

We present *explicit* Horn-geometric axiomatization for the theory of Faithfully Quadratic Rings, answering a question posed in [**DM3**].

Keywords: Quadratic Forms, Special Groups, Faithfully Quadratic Rings, Horn-Geometric sentences

Introduction

The classical theory of algebraic quadratic forms over fields of characteristic $\neq 2$ started in the late 1930's with E. Witt and was put foward in the late 1960's by A. Pfister. In the decade of 1980's, appeared the first abstract approach to quadratic forms theory as Marshall's abstract space of orderings ([**Mar**]). Only in early 1990's that arise a (finitary) first-order theory that generalizes, simultaneously, the reduced and non-reduced theory of quadratic forms called special groups ([**DM1**] and Definition 1).

In [**DM3**], the theory of Special Groups is employed to generalize results on quadratic forms over fields to a wide class of (commutative, unitary) rings, therein named faithfully quadratic rings, that includes rings with many units, reduced f-rings (herein rings of continuous real-valued functions) and strictly representable rings. If A is a T-faithfully quadratic ring and $G_T(A)$ is its associated special group, then T-isometry and T-representation in the ring A are faithfully encoded by isometry and representation in $G_T(A)$. In fact, the theory can be extended to pairs (S, T) where T is a quadratic set and S is a T-subgroup (cf. Definition 6 below), which generalize Definition 2.2, p.10, of [**DM3**], and gives a unified treatment to the subject.

By a combination of results in [**DM3**], the first-order theory of T-faithfully quadra tic rings is *both* Horn and geometric. By Corollary 5.6, p.58, in [**DM3**], the theory of T-faithfully quadratic rings has a Horn axiomatization -which involves a

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non-constructive model-theoretic result of Keisler, Galvin and Shelah ([**CK**], Theorem 6.2.5)- while Theorem 5.2, p. 54, in [**DM3**] shows that T-faithfully quadratic rings is a geometric theory. The question posed is if *explicit* Horn-geometric axioms could be provided for that theory (cf. paragraph right after the proof of Corollary 5.6, p. 58). It should be remarked that Horn-geometrical axioms for faithfully quadratic rings, in the case were T is the set of all squares of the ring, appears in Theorem 5.2.(a), p. 54, of [**DM3**].

In this work we provide a complete and explicit list of Horn-geometric axioms for T-quadratic faithfullness (Theorem 19). In the case where T is not necessarily the set of all squares of the ring, the contribution of our short work is the equivalence between T-isometry (Definition 2.17, p. 18, [**DM3**]) and the notion of T-congruence, introduced in Definition 12 of the paper, allowing the elimination of the disjunction appearing in underlined formula in p.56 of [**DM3**] and yielding a Horn-geometric axiomatic for the theory of Faithfully Quadratic Rings.

In all that follows: the word "ring" stands for a commutative unitary ring A, wherein 2 is a unit.

1 Preliminaries

Let A be a set and $\equiv$ a binary relation on A^2. We extend $\equiv$ to A^n, $n \geq 3$, by induction as follows: given $a_1, \ldots, a_n, b_1, \ldots, b_n \in A$, we say that $\langle a_1, \ldots, a_n \rangle \equiv_n \langle b_1, \ldots, b_n \rangle$ if there are $\alpha, \beta, z_3, \ldots, z_n \in A$ such that

$$\begin{gathered}\langle a_1, \alpha \rangle \equiv \langle b_1, \beta \rangle \\ \langle a_2, \ldots, a_n \rangle \equiv_{n-1} \langle \alpha, z_3, \ldots, z_n \rangle \\ \langle b_2, \ldots, b_n \rangle \equiv_{n-1} \langle \beta, z_3, \ldots, z_n \rangle.\end{gathered}$$

For $n = 1$, we adopt the convention $\langle a \rangle \equiv_1 \langle b \rangle \Leftrightarrow a = b$. In general, the subscript of $\equiv_n$ is omitted and we use just $\equiv$. This extension of $\equiv$ comes from the inductive characterization of the quadratic forms isometry relation over fields of characteristic $\neq 2$.

Definition 1 (Def. 1.9, [**DM3**]). *A **proto-special group** (abbreviation: π-SG) is triple $(G, -1, \equiv_G)$ where G is a group of exponent 2, $-1 \in G$ (denote: $-x$ for $-1 \cdot x$, $\forall x \in G$) and $\equiv_G$ is a binary relation on $G \times G$ satisfying for all $a, b, c, d, x \in G$:*

SG-0 *$\equiv_G$ is an equivalence relation.*

SG-1 $\langle a, b \rangle \equiv_G \langle b, a \rangle$.

SG-2 $\langle a, -a\rangle \equiv_G \langle 1, -1\rangle$.

SG-3 $\langle a, b\rangle \equiv_G \langle c, d\rangle \Rightarrow ab = cd$.

SG-5 $\langle a, b\rangle \equiv_G \langle c, d\rangle \Rightarrow \langle xa, xb\rangle \equiv_G \langle xc, xd\rangle$.

*A π-SG is called **reduced (π-RSG)** if it satisfies* **red** *below and $-1 \neq 1$ (i.e. it is formally real).*

red $\langle a, a\rangle \equiv_G \langle 1, 1\rangle \Rightarrow a = 1$.

*Furthermore, a π-SG $(G, \equiv_G, -1)$ is called **pre-special group (pSG)** if for all $a, b, c, d \in G$*

SG-4 $\langle a, b\rangle \equiv_G \langle c, d\rangle \Rightarrow \langle a, -c\rangle \equiv_G \langle -b, d\rangle$

*and is a **special group (SG)** if satisfies $SG-4$ and*

SG-6 *The extension of $\equiv_G$ to G^3 is a transitive relation.*

Remark 1. *i) The axiom **SG-6** of Special Group definition seems to be very specific but it implies that for all $n \geq 1$, the extended relation $\equiv_n$ is an equivalence relation. See Theorem 1.23 of [**DM1**] for a proof of this fact.*

ii) A n-form over G is a tuple $\varphi = \langle a_1, \ldots, a_n\rangle$, where $n \in \mathbb{N}$ and $a_1, \ldots, a_n \in G$.

Definition 2 (Def. 1.3, [**DM1**])**.** *Let $(G, \equiv, -1)$ a π-SG and $\varphi = \langle a_1, \ldots, a_n\rangle$ a form over G. The set of elements represented by φ is*

$$D_G(\varphi) = \{x \in G\colon \text{there are } b_2, \ldots, b_n \in G \text{ such that } \langle x, b_2, \ldots, b_n\rangle \equiv \varphi\}.$$

Example 3.
- *Let K be a field of characteristic $\neq 2$. Consider $G(K) = K^\times/K^{\times 2}$, where $K^\times = K \setminus \{0\}$ is the set of all units of K, and the binary relation in $G(K)$ given by $\langle \overline{a}, \overline{b}\rangle \equiv \langle \overline{c}, \overline{d}\rangle$ iff $\overline{ab} = \overline{cd}$ in $G(K)$ and $a = cx^2 + dy^2$ for some $x, y \in \dot{K}$. Then $(G(K), \equiv, \overline{-1})$ is a special group (Theorem 1.32 of [**DM1**]).*
- *The above construction can be generalized to a broad class of rings that satisfies the conditions of Definition 15 below: this includes the rings with many units, reduced f-rings (herein rings of continuous real-valued functions) and strictly representable rings. For more details, see [**DM3**].*

Definition 4 (Def. 1.4, [**DM3**])**.** *Let $\mathcal{L}$ a first-order language.*

*i) A formula φ is **positive-primitive (pp-formula)** if it is equivalent to $\exists\overline{v}\psi(\overline{v})$ where $\psi(\overline{v})$ is a conjunction of atomic formulas.*

*ii) A formula φ is called **Horn-geometrical** in the free variables $\overline{t}$ if it is the negation of an atomic formula or of the form $\forall\overline{v}\psi_1(\overline{v},\overline{t}) \to \psi_2(\overline{y},\overline{t})$ where ψ_1, ψ_2 are pp-formulas.*

Example 5.

- *The theory of commutative unitary rings is Horn-geometrical.*
- *The theory of (reduced) special groups is Horn-geometrical.*

The next definition sets the grounds to the theory of faithfull quadratic rings and slightly generalize the Definition 2.2, p.10, of [**DM3**], introducing the (unifying) notion of quadratic set.

Definition 6. *Let A be a ring.*

- *A subset $T \subseteq A$ is called **quadratic** if it is multiplicative (i.e., it closed under the ring product) and $A^2 \subseteq T$.*
- *Given a quadratic set $T \subseteq A$, a set $S \subseteq A^\times$ is called T**-subgroup** if it is multiplicative and $T^\times \cup \{-1\} \subseteq S$. When $T = A^2$, the A^2-subgroups are called q-subgroups.*

Let $T \subseteq A$ be a quadratic set and $S \subseteq A^\times$ be a T-subgroup. Given $a, b \in S$, the valued-represented mod T (p. 11, [**DM3**]) by $\langle a, b\rangle$ is

$$D^v_{S,T}(a,b) = \{c \in S \colon \text{there are } t, t' \in T \text{ such that } c = at + bt'\}.$$

Moreover, if $D^v_{S,T}(1,a)$ is multiplicative, for every $a \in S$, then (S,T) is called a **quadratic pair**.

Remark 2. *Let A be a ring.*

- *If $T = A^2$ or T is a preorder in A (i.e., if $A^2 \subseteq T, T \cdot T \subseteq T, T + T \subseteq T$), then T is a quadratic set in A.*

  Let T be a quadratic set in the ring A.
- *The smallest T-subgroup is $T^\times \cup -T^\times$ and the largest is $A^\times$.*
- *If S is a T-subgroup, then S is a subgroup of $A^\times$.*
- *Since $0, 1 \in T$, $\{a, b\} \subseteq D^v_{S,T}(a,b)$.*

Remark 3. *Let A be a ring, T be a quadratic set in A and S be a T-subgroup.*

- *If (S,T) is a quadratic pair, then $D^v_{S,T}(1,a) \subseteq S$ is a subgroup for all $a \in S$. In fact, $1 \in D^v_{S,T}(1,a)$ and given $x \in D^v_{S,T}(1,a)$, let $u,v \in T$ such that $x = u + av$. Then $x^{-1} = x^{-2}x = ux^{-2} + avx^{-2} \in D^v_{S,T}(1,a)$. Since $D^v_{S,T}(1,a)$ is multiplicative, it follows that $D^v_{S,T}(1,a) \subseteq S$ is a subgroup.*

- *If (S,T) is a quadratic pair and $S' \subseteq S$ is another T-subgroup, then (S',T) is also a quadratic pair. In particular, if $(A^\times,T)$ is a quadratic pair, then every T-subgroup determines a quadratic pair.*

- *A very interesting class of examples of quadratic pairs (S,T) is given in Theorem 2.3 in [**DM2**]: they are able to represent all reduced special groups i.e., for each special group G, there exists a quadratic pair (S,T) such that $G \cong G_T(S)$. In more details: $A_G := C(X_G,\mathbb{R})$, where X_G is the (boolean) space of all SG-morphisms $G \to \{-1,1\}$ and $T = A_G^2$.*

In the field case, there is a useful criteria to determine a SG-subgroup (Definition 1.28 and Lemma 1.29, p.21, [**DM1**]). In the ring case, an analogous strategy works to obtain quadratic pairs.

Proposition 7. *Let A be a ring, $T \subseteq A$ quadratic set and $S \subseteq A^\times$ a T-subgroup. Suppose that for all $p,q,u,v \in T$ and $a \in S$, exists $x \in A$ such that*

$$(pua^2 + qv - xa) \in T \text{ and } (pv + qu + x) \in T. \tag{*}$$

Then (S,T) is a quadratic pair. In particular, if $T = A^2$ or T is a preorder, then (S,T) is a quadratic pair.

Proof. Let $s,t \in D^v_{S,T}(1,a)$, $a \in A$. Take $p,q,u,v \in T$ with

$$s = pa + q \text{ and } t = ua + v.$$

Then $st = (pua^2 + qv) + (pv + qu)a$. By hypothesis, exists $x \in A$ such that $(pua^2 + qv - xa) \in T$ and $(pv + qu + x) \in T$. Then

$$st = (pua^2 + qv - xa) + (pv + qu + x)a \in D^v_{S,T}(1,a).$$

If T is closed under sums (preorder), then $(*)$ is satisfied with $x = 0$ for all $a \in A$. If $T = A^2$, then $p = p_1^2, q = q_1^2, u = u_1^2, v = v_1^2$; if we take $x = 2(p_1q_1u_1v_1)$, then $(pua^2 + qv - xa) = (p_1u_1a - q_1v_1)^2 \in T$ and $(pv + qu + x) = (p_1v_1 + q_1u_1)^2 \in T$. $\square$

The next lemma is a version of Lemma 2.6, p.11, [**DM3**] for quadratic pairs and gives some properties for the valued-represented set of a 2-form. Note that the proofs follows without any major change.

Lemma 8. *Let A be a ring and (S,T) a quadratic pair. Let $x,y,u,v \in S$ and $t,w \in T^\times$.*

i) $uD^v_{S,T}(x,y) = D^v_{S,T}(ux,uy)$ *and* $D^v_{S,T}(x,y) = D^v_{S,T}(tx,wy)$.

ii) $u \in D^v_{S,T}(x,y) \Rightarrow ut \in D^v_{S,T}(x,y)$.

iii) $x \in D^v_{S,T}(1,y) \Rightarrow D^v_{S,T}(x,xy) = xD^v_{S,T}(1,y) = D^v_{S,T}(1,y)$.

iv) $u \in D^v_{S,T}(x,y) \Leftrightarrow D^v_{S,T}(u,uxy) = D^v_{S,T}(x,y)$.

v) Assuming $xy = uvt$, the following are equivalent:

(a) $D^v_{S,T}(x,y) = D^v_{S,T}(u,v)$*;*

(b) $D^v_{S,T}(x,y) \cap D^v_{S,T}(u,v) \neq \emptyset$

Proof. Items i) and ii) are straightforward. For the statement iii), the first equality is a direct consequence of i). For the last, since (S,T) is a quadratic pair, the set $D^v_{S,T}(1,a) \subseteq S$ is a subgroup (Remark 3) and thus

$$xD^v_{S,T}(1,a) \subseteq D^v_{S,T}(1,a) \text{ and } x^{-1}D^v_{S,T}(1,a) \subseteq D^v_{S,T}(1,a),$$

which establishes the equality.

iv) Since $u \in D^v_{S,T}(u,uxy)$, the direction $\Leftarrow$ is immediate. For the reverse, note that if $u \in D^v_{S,T}(x,y)$, then by item i) we have $ux \in D^v_{S,T}(x^2,xy) = D^v_{S,T}(1,xy)$ and thus by item ii) one can see that $D^v_{S,T}(ux,ux^2y) = D^v_{S,T}(1,xy)$. Multiplying this equation by x we obtain

$$D^v_{S,T}(u,uxy) = xD^v_{S,T}(ux,ux^2y) = xD^v_{S,T}(1,xy) = D^v_{S,T}(x,y).$$

v) The direction $a) \Rightarrow b)$ is immediate. For the other, assume that $z \in D^v_{S,T}(x,y) \cap D^v_{S,T}(u,v)$. Then by item iv) one has

$$D^v_{S,T}(x,y) = D^v_{S,T}(z,zxy) = D^v_{S,T}(z,zuvt) = D^v_{S,T}(z,zuv) = D^v_{S,T}(u,v).$$

$\square$

Following the Chapter 2 of [**DM3**], fixed a quadratic pair (S,T), a candidate for special group is associated to the pair (S,T) in the following way:

- The group is given by $G_T(S) = S/T^\times$. Since $A^2 \subseteq T$, the group $G_T(S)$ has exponent 2. Furthermore, the distinguished element in $G_T(S)$ is $\overline{-1}$ (in general, denoted just by -1).
- For $\overline{a}, \overline{b}, \overline{c}, \overline{d} \in G_T(S)$, the 2-isometry is given by

$$\langle \overline{a}, \overline{b} \rangle \equiv_{S,T} \langle \overline{c}, \overline{d} \rangle \Leftrightarrow \overline{a}\overline{b} = \overline{c}\overline{d} \text{ and } D^v_{S,T}(a,b) = D^v_{S,T}(c,d).$$

  Note that by Lemma 8.(i), the last equality of valued-represented sets it does not dependent of a particular choice of members in the equivalence classes.

The main properties of $G_T(S)$ are summarized below.

Proposition 9. *Let A be a ring and (S,T) a quadratic pair.*

i) The triple $(G_T(S), \equiv_{S,T}, -1)$ is a π-SG. Furthermore, for $x,y,z \in S$,

$$z \in D^v_{S,T}(x,y) \Leftrightarrow \overline{z} \in D_{G_T(S)}(\overline{x}, \overline{y}).$$

*ii) If (S,T) is **2-transversal**, that is, for all $x,y \in S$,*

$$D^v_{S,T}(x,y) = \{c \in S\colon \text{there are } u,v \in T^\times : c = ux + vy\},$$

then $G_T(S)$ is a pre-special group.

iii) $G_T(S)$ is reduced π-SG if, and only if, $-1 \notin T$ and $(T+T) \cap S \subseteq T$. In particular, if T is a proper preorder, then $G_T(S)$ is reduced.

Proof. The proof follows by very similar arguments of Lemma 2.7, p. 13, [**DM3**], which consists in an application of Lemma 8. □

2 Main Result

From now on, (S,T) will denote a quadratic pair in a ring A, unless explicitly stated otherwise. A tuple $\varphi = \langle a_1, ..., a_n \rangle$, $a_i \in S$, is called a n-form over S; also let $M(\varphi) = M((a_i)_{i \leq n})$ be the $n \times n$ diagonal matrix whose non-zero elements are precisely the diagonal with $a_1, ..., a_n$, in this order. The notion of T-isometry appears in Definition 2.17, p. 18, [**DM3**]. The definition of T-congruence, to be shown equivalent to T-isometry, is introduced in order to yield a Horn-geometric axiomatization of the theory of T-faithfully quadratic rings.

Definition 10. *Let $\varphi = \langle a_1, ..., a_n\rangle$ and $\psi = \langle b_1, ..., b_n\rangle$ be two n-forms over S.*

- *The forms φ, ψ are **simply T-isometric**, written $\varphi \approx^*_{S,T} \psi$, if there are $t_1, ..., t_n \in T^\times$ such that $b_i = t_i a_i$ or $N \in GL_n(A)$ such that $NM(\varphi)N^t = M(\psi)$.*

- *φ and ψ are **simply T-congruent**, written $\varphi \cong^*_{S,T} \psi$, if there are $N \in GL_n(A)$ and $x_1, ..., x_n, y_1, ..., y_n \in T^\times$ such that*

$$NM((x_i)_{i\leq n})M(\varphi)N^t = M((y_i)_{i\leq n})M(\psi).$$

Proposition 11. *The relations $\approx^*_{S,T}$ (simply T-isometry) and $\cong^*_{S,T}$ (simply T-congruence) over the set of n-forms, $n \geq 1$, are reflexive and symmetric.*

Proof. Let $\varphi = \langle a_1, \ldots, a_n\rangle, \psi = \langle b_1, \ldots, b_n\rangle$ be forms over S.

- Reflexivity: Since $a_i = 1 \cdot a_i$ for all $i = 1, \ldots, n$, $\varphi \approx^*_{S,T} \varphi$.

  Similarly, we have $Id_n(A)Id_n(A)M(\varphi)Id_n(A)^t = Id_n(A)M(\varphi)$ and so $\varphi \cong^*_{S,T} \varphi$.

- Symmetry: Assume that $\varphi \approx^*_{S,T} \psi$. If there are $t_1, \ldots, t_n \in T^\times$ with $b_i = t_i a_i$, then $a_i = t_i^{-1} b_i$ and $t_i^{-1} = t_i^{-2} t_i \in T^\times$; so $\psi \approx^*_{S,T} \varphi$. If there is $N \in GL_n(A)$ such that $NM(\varphi)N^t = M(\psi)$, then $P = N^{-1}$ satisfies $PM(\psi)P^t = M(\varphi)$; so $\psi \approx^*_{S,T} \varphi$.

  Now suppose that $\varphi \cong^*_{S,T} \psi$. Then there are $x_i, y_i \in T^\times$, $i = 1, \ldots, n$, and $N \in GL_n(A)$ such that $NM((x_i)_{i\leq n})M(\varphi)N^t = M((y_i)_{i\leq n})M(\psi)$. Thus, $P = N^{-1}$ satisfies $PM((y_i)_{i\leq n})M(\psi)P^t = M((x_i)_{i\leq n})M(\varphi)$ and therefore $\psi \cong^*_{S,T} \varphi$.

$\square$

Definition 12. *Let φ, ψ be two n-forms, $n \geq 1$, over S.*

- *The forms φ, ψ are said to be T-isometric, written $\varphi \approx_{S,T} \psi$, if there are n-forms $\varphi_0, \ldots, \varphi_k$ such that $\varphi_0 = \varphi, \varphi_k = \psi$ and $\varphi_i \approx^*_{S,T} \varphi_{i+1}$ for all $i = 0, \ldots, k-1$.*

- *The forms φ, ψ are T-congruent, written $\varphi \cong_{S,T} \psi$, if there are n-forms $\varphi_0, \ldots, \varphi_k$ such that $\varphi_0 = \varphi, \varphi_k = \psi$ and $\varphi_i \cong^*_{S,T} \varphi_{i+1}$ for all $i = 0, \ldots, k-1$.*

Note that the relations $\approx_{S,T}$ and $\cong_{S,T}$ are, respectively, the transitive closure of the relations $\approx^*_{S,T}$ and $\cong^*_{S,T}$ thus, by Proposition 11, they are equivalence relations on the set of n-forms over S.

The following (easily established) result is fundamental to obtain our main contribution.

Proposition 13. *Let $\varphi = \langle a_1, ..., a_n\rangle$ and $\psi = \langle b_1, ..., b_n\rangle$ be two n-forms over S. Then $\varphi \approx_{S,T} \psi$ iff $\varphi \cong_{S,T} \psi$.*

Proof. It's enough to show that 1) $\varphi \approx^*_{S,T} \psi \Rightarrow \varphi \cong_{S,T} \psi$ and 2) $\varphi \cong^*_{S,T} \psi \Rightarrow \varphi \approx_{S,T} \psi$.
1) Follows easily from definitions.
2) Assuming $\varphi \cong^*_{S,T} \psi$, take $x_1, ..., x_n, y_1, ..., y_n \in T^\times$ and $N \in GL_n(A)$ such that $NM((x_i)_{i\leq n})M(\varphi)N^t = M((y_i)_{i\leq n})M(\psi)$. Then we have

$$\varphi \approx^*_{S,T} \langle x_1a_1, ..., x_na_n\rangle \approx^*_{S,T} \langle y_1b_1, ..., y_nb_n\rangle \approx^*_{S,T} \psi.$$

□

Definition 14 (Def. 2.24, [**DM3**])**.** *Let A be a ring and (S,T) a quadratic pair. Let S be T-subgroup. Let $\varphi = \langle a_1, \dots, a_n\rangle$ be a form over S and $\overline{\varphi} = \langle \overline{a_1}, \dots, \overline{a_n}\rangle$ be the corresponding form in $G_T(S)$.*

a) $D_{S,T}(\varphi) = \{x \in S\colon$ there are $b_2, \dots, b_n \in S$ such that $\overline{\varphi} \equiv_{G_T(S)} \langle \overline{x}, \overline{b_2}, \dots, \overline{b_n}\rangle\}$ is the set of elements of S **isometry-represented** *mod T by φ in $G_T(S)$.*

b) $D^v_{S,T}(\varphi) = \{x \in S\colon$ there are $x_1, \dots, x_n \in T$ such that $x = \sum_{i=1}^n x_ia_i\}$ is the set of elements of S **value-represented** *mod T by φ.*

c) $D^t_{S,T}(\varphi) = \{x \in S\colon$ there are $x_1, \dots, x_n \in T^\times$ such that $x = \sum_{i=1}^n x_ia_i\}$ is the set of elements of S **transversely represented** *mod T by φ.*

d) The set $\mathcal{D}_{S,T}(\varphi)$ of elements of S **inductively represented** *mod T by φ is defined as follows:*

- *If $n = 1$, then $\mathcal{D}_{S,T}(\varphi) = \{a_1\}$;*
- *If $n = 2$, then $\mathcal{D}_{S,T}(\varphi) = D^v_{S,T}(\varphi)$;*
- *If $n \geq 3$, then*

$$\mathcal{D}_{S,T}(\varphi) = \bigcap_{k=1}^{n} \bigcup \{D^v_{S,T}(a_k, u)\colon u \in D^v_{S,T}(a_1, \dots, \check{a_k}, \dots, a_n)\}.$$

Remark 4. *Consider the notation from Definition* 14. *For all forms φ over S, the following is satisfied without any further assumptions:*

- $D_{S,T}(\varphi) \subseteq D^v_{S,T}(\varphi)$;
- $D^t_{S,T}(\varphi) \subseteq D^v_{S,T}(\varphi)$;
- $\mathcal{D}_{S,T}(\varphi) \subseteq D^v_{S,T}(\varphi)$;
- *If* $dim(\varphi) \leq 3$, *then* $\mathcal{D}_{S,T}(\varphi) \subseteq D_{S,T}(\varphi)$.

*If $T = A^2$ or T is pre-order, Lemma 2.26 in [**DM3**], contains a proof of these statements. An analogous reasoning establishes the inclusions in the general case of quadratic pairs.*

Definition 15 (Def. 3.1, [**DM3**])**.** *Let A be a ring and (S,T) a quadratic pair. The T-subgroup S is called T**-faithfully quadratic** if the following are satisfied:*

i) For all $a, b \in S$,

$$D^v_{S,T}(a,b) = D^t_{S,T}(a,b);$$

ii) For all form φ with dimension $n \geq 2$,

$$\mathcal{D}_{S,T}(\varphi) = D^v_{S,T}(\varphi);$$

iii) For all forms φ, ψ of same dimension and for all $a \in S$

$$\langle a \rangle \oplus \varphi \approx_{S,T} \langle a \rangle \oplus \psi \Rightarrow \varphi \approx_{S,T} \psi.$$

Definition 16 ([**DM3**], p. 27)**.** *Let A be a ring and $T \subseteq A$ a quadratic set. The ring A is called T-faithfully quadratic ring if*

- *The pair $(A^\times, T)$ is quadratic;*
- *The T-subgroup $A^\times$ is T-faithfully quadratic.*

In the case $T = A^2$, a A^2-faithfully quadratic is simply called faithfully quadratic ring.

The next result illustrates the potential of T-faithfully quadratic notion in provide a special group that faithfully represent the T-isometry and T-representation of a ring.

Theorem 17. *Let A be a ring and (S,T) a quadratic pair. Assume that S is T-faithfully quadratic. Let φ,ψ be forms over S of same dimension.*

i) $D^v_{S,T}(\varphi) = D^t_{S,T}(\varphi)$, *that is, an element in S is valued-represented iff it is transversally represented.*

ii) $D^v_{S,T}(\varphi) = D_{S,T}(\varphi)$, *that is, the set of valued-represented elements coincide with those isometry-represented in $G_T(S)$.*

iii) *If $\varphi = \varphi_1 \oplus \cdots \oplus \varphi_k$, then*

$$D^v_{S,T}(\varphi) = \bigcup \{D^v_{S,T}(u_1,\ldots,u_k)\colon u_i \in D^v_{S,T}(\varphi_i), i = 1,\ldots,k\}.$$

iv) $\varphi \approx_{S,T} \psi$ *if, and only if,* $\overline{\varphi} \equiv_{G_T(S)} \overline{\psi}$.

v) $(G_T(S), \equiv_{G_T(S)}, -1)$ *is a special group.*

Proof. The proof in the case $T = A^2$ or T is a preorder can be found in Theorems 3.3, 3.5 and 3.6 of [**DM3**] with emphasis in which axioms of faithfully quadratic pairs are needed to obtain each of above results. Very similar arguments entails the proof in for quadratic pairs. In 3.7, [**DM3**], there is a fruitful discussion about the reason and utility of faithfully quadratic rings to obtain a rich theory of quadratic forms over invertible coefficients. □

Proposition 18. *Let A be a ring and (S,T) a quadratic pair. Assume that S is T-faithfully quadratic. Given two forms φ and ψ of same dimension $n \geq 2$ such that $\varphi \cong_{S,T} \psi$, then it is possible to find $\varphi_0,...,\varphi_{l(n)}$, $l(n) = 2^{n-1} - 1$, $\varphi_0 = \varphi, \varphi_{l(n)} = \psi$ such that $\varphi_i \cong^*_{S,T} \varphi_{i+1}$ for every $i < l(n)$.*

Proof. This can be obtained from a simple adaptation of the proof of Proposition 5.1, p.53, [**DM3**], combined with the Proposition 13. The proof of Lemma 2.21, p.20, [**DM3**], shows that $l(2) = 1$. The remaining of the proof follows essentially from Theorem 17.(iv) together with the inductive description of isometry in special groups (described before Definition 1). □

In Theorem 5.2, p.54, [**DM3**] is proved that the theory of faithfully quadratic rings is Horn-geometric and in Remark 5.3 of [**DM3**] is discussed how to adapt the proof of Theorem 5.2 to obtain a Horn-geometric axiomatization to faithfully quadratic q-subgroup. This result can be generalized to pre-orders and even for quadratic pairs. This is the content of next theorem.

Theorem 19. *The theory of quadratic pairs (S,T) such that S is T-faithfully quadratic is Horn-Geometrical in the language $L=\{+,\cdot,0,1,-1,S,T\}$, where S,T are unitary relational symbols.*

Proof. First of all, note that when (S,T) is a quadratic pair, $S\cap T=T^\times$. So we will use the formula $T^\times(a)$ as an abbreviation to the pp-formula $S(a)\wedge T(a)$. To simplify the formulas below, we should also note that $a\in A^\times$ is equivalent to $T^\times(a^2)$ and so $a\in A^\times$ is also a conjunction of atomic formulas. With this, for $n\geq 2$, we define:

- Let $(\phi_n)^v_{S,T}(a,b_1,...,b_n)$ be the formula in $n+1$ free variables given by

$$\exists t_1,...,t_n(\bigwedge_{i=1}^n(S(b_i)\wedge T(t_i))\ \wedge\ S(a)\ \wedge\ a=\sum_{i=1}^n t_ib_i).$$

  Note that this formula is a pp-formula and it express $a\in D^v_{S,T}(b_1,...,b_n)$.

- Let $(\phi_n)^t_{S,T}(a,b_1,...,b_n)$ be the formula in $n+1$ free variables given by

$$\exists t_1,...,t_n(\bigwedge_{i=1}^n(S(b_i)\wedge T^\times(t_i))\ \wedge\ S(a)\ \wedge\ a=\sum_{i=1}^n t_ib_i).$$

  Note that this formula is a pp-formula and it encodes $a\in D^t_{S,T}(b_1,...,b_n)$.

  Thus, fixed $n\geq 2$, let $l=l(n)$ from Proposition 18. Consider $(\psi_n)_{S,T}(\overline{a},\overline{b})$ to be the formula in $2l$ variables given by

$$\exists\overline{c_0},...,\overline{c_l},N_1,...,N_l,\overline{x_1},...,\overline{x_l},\overline{y_1},...,\overline{y_l}$$
$$(\bigwedge_{i=1}^l(\det(N_i)\in A^{\times 1})\ \wedge\ \bigwedge_{i=1}^l(T^\times(\overline{x_i})\wedge T^\times(\overline{y_i})\wedge S(\overline{c_i}))\ \wedge\ \overline{c_0}=\overline{a}\ \wedge\ \overline{c_l}=\overline{b}$$
$$\wedge\ \bigwedge_{i=1}^l(N_i\cdot M((\overline{x_i})_{i\leq n})\cdot M((\overline{c_{i-1}})_{i\leq n})\cdot N_i^t=M((\overline{y_i})_{i\leq n})\cdot M((\overline{c_i})_{i\leq n}))$$

  This formula is a pp-formula and it express $\langle a_1,...,a_n\rangle\cong_{S,T}\langle b_1,...,b_n\rangle$.

Now, a possible set of axioms to the theory of faithfully quadratic rings can be given by

[1] Note that the phrase "the determinant of N_i is an invertible element of A" can be encoded by a pp-formula in the language L.

i) axioms ensuring that $\{A,+,\cdot,0,1,-1\}$ is a commutative ring with unity with $2 \in A^{\times}$;

ii) $\forall a,b\ (T(a)\ \wedge\ T(b) \Rightarrow T(ab))$;

iii) $\forall a\ (T(a^2))$;

iv) $\forall a\ (S(a) \Rightarrow \exists u\ au = 1)$;

v) $\forall a,b\ (S(a)\ \wedge\ S(b) \Rightarrow S(ab))$;

vi) $\forall a,u\ (T(a)\ \wedge\ au = 1 \Rightarrow S(a))$;

vii) $S(-1)$;

viii) $\forall a,x,y\ ((\phi_2)^v_{S,T}(x,1,a) \wedge (\phi_2)^v_{S,T}(y,1,a) \Rightarrow (\phi_2)^v_{S,T}(xy,1,a))$;

Until now the axioms basically express that (S,T) is a quadratic pair in the ring A with $2 \in A^{\times}$. Now we describe the axioms for S to be T-faithfully quadratic:

ix) $\forall a,b,x\ ((\phi_2)^v_{S,T}(x,a,b) \Rightarrow (\phi_2)^t_{S,T}(x,a,b))$;

For each $n \geq 2$,

x) $\forall a_1,...,a_n,x\ ((\phi_n)^v_{S,T}(x,a_1,...,a_n)\ \Rightarrow\ \bigwedge_{i=1}^{n} \exists u\ ((\phi_{n-1})^v_{S,T}(u,a_1,...,\check{a}_i,...,a_n)\ \wedge\ (\phi_2)^v_{S,T}(x,a_i,u)))$;

xi) $\forall a,a_1,...,a_n,b_1,...,b_n\ ((\psi_{n+1})_{S,T}(a,\overline{a},a,\overline{b}) \Rightarrow (\psi_n)_{S,T}(\overline{a},\overline{b}))$.

Let $\mathcal{T}$ be the theory determined by the sentences including in the itens $i) - xi)$ above. Note that $\mathcal{T}$ is a Horn-geometric theory. An argument analogous to that given in the end of the proof of Theorem 5.2, p. 56, in [**DM3**], will show that a $\langle A,S,T\rangle$ is a model of $\mathcal{T}$ iff A is a ring with $2 \in A^{\times}$, (S,T) is quadratic pair and S is T-faithfully quadratic. Furthermore, note that it is possible to require additionally that $\overline{1} \neq \overline{-1}$ in $G_T(S)$ by adding the axiom $\neg T(-1)$, which is yet a Horn-geometric formula. □

Remark 5. *In the above theorem, if the T-subgroup S can be described by a pp-sentence in the language $\{+,\cdot,0,1,-1,T\}$, then minor adaptations in the proof also yields a Horn-geometric theory without the symbol S in the language. This is the case when $S = A^{\times}$ or $S = T^{\times} \cup -T^{\times}$.*

References

[**CK**] C.C. Chang, H.J. Keisler, **Model Theory**, North-Holland Publ. CO., Amsterdam, 1990.

[**DM1**] M. Dickmann, F. Miraglia, **Special Groups: Boolean-Theoretic Methods in the Theory of Quadratic Forms**, Memoirs of the AMS **689**, American Mathematical Society, Providence, USA, 2000.

[**DM2**] M. Dickmann, F. Miraglia, *Representation of reduced special groups in algebras of continuous functions*, Quadratic forms—algebra, arithmetic, and geometry, Contemp. Math., vol. 493, Amer. Math. Soc., Providence, RI, 2009, pp. 83–97.

[**DM3**] M. Dickmann, F. Miraglia, **Faithfully Quadratic Rings**, Memoirs of AMS **1128**, American Mathematical Society, Providence, USA, 2015.

[**Mar**] M. A. Marshall, **Spaces of Orderings and Abstract Real Spectra**, Lecture Notes in Mathematics **1636**, Springer-Verlag, Berlin, Germany, 1996.

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